Arjun Juneja

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ImArjunJ

Summary

Low-level systems engineer with a preprint research paper (arXiv) and 7 first-place hackathon wins across Europe and the US. I specialize in kernel development, reverse engineering, and high-performance native applications in C++ and Rust.

Experience

XTX Markets *Security Engineer* **London, UK** *Jul 2026 – Sep 2026*
- Incoming Summer 2026. Security tooling and infrastructure for a quantitative trading firm.

Hacktron AI *Software Engineer* **London, UK** *Sep 2025 – Jan 2026*
- Built AI-driven callstack generation engine using treesitter AST parsing across 6 languages for automated pentesting reports
- Designed and shipped CLI authentication flow handling OAuth2 token exchange and session persistence

BP *Software Engineer Intern* **London, UK** *Jul 2023 – Sep 2023*
- Contract extended 2 months. Built Safety & Assurance panel adopted by 100+ assurers across the organisation

Projects & Publications

RX-INT: Kernel-Assisted Security Tool
Windows kernel driver for real-time in-memory threat detection (VAD manipulation, module stomping). Outperformed PE-sieve across all tested stealth injection techniques (12 scenarios). Preprint on arXiv. Paper

jcan
Open-source CAN bus diagnostic tool (83+ stars). First-ever Linux driver for Vector CAN hardware. Features signal plotting, real-time bus monitor, DBC decoding, and multi-adapter support. C++23, Dear ImGui, libusb. Project

Chimera: .NET Runtime Injection
Stealth process injection via CLR JIT internals. Reverse-engineered the .NET runtime to force RWX allocations through internal JIT code heap paths, indistinguishable from legitimate JIT regions. Writeup

LogiLinux
1st place, LauzHack '25. Reverse-engineered Logitech HID++ protocols to build full Linux driver for MX Creative Console. C++ backend, Tauri UI, plugin SDK. Project

Source2 Engine RE
Reverse-engineered Valve's Source2 engine (CS2). Documented runtime interface system, virtual method tables, and memory layout. Built tooling for interface enumeration and SDK generation. Project Writeup

osu! RE
Reverse-engineered osu!'s .NET runtime structure, and memory layout. Built pattern scanner and real-time memory reader for audio timing and game state extraction. Writeup

Competitions

- **1st Place**: pwnEd 7 CTF (pwn/rev, 2026), Junction '25 (€10k), LauzHack '25 (EPFL), DevHouse SF ($1k), SotonHack '25, JumpStart '24
- **Top Placements**: ICHack '26 (3rd + Best Technical), RoboHack '25 (2nd), ICHack '25 (4th)
- **Meta Hacker Cup**: Top 6.5% (1465 / 22,494) Certificate

Education

University of Southampton *BSc Computer Science* **Southampton, UK** *Sep 2024 – Jul 2027*
Year 1: First Class

Feltham College *A-Levels: Mathematics, Physics, Chemistry* **London, UK** *Sep 2022 – Jul 2024*
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Technical Skills

Core: C++, Rust, C#/.NET, Python, LLVM IR, x86/x64 Assembly
Systems: Windows Kernel, Win32 API, Linux, FPGA, System Verilog, HID, CAN bus
Tools: IDA Pro, Ghidra, x64dbg, Wireshark, GDB, CMake, Docker, Git, CI/CD
Graphics: OpenGL, Dear ImGui, Compute Shaders